

Challenge 1: Autonomous Content-to-Action Agent (Insight → Action System)

Challenge Overview

In today's world, organizations are flooded with unstructured information — reports, news, dashboards, policy updates — but most AI systems stop at **summarization or analysis**.

Real-world systems must go further.

They must:

- understand information
- extract insights
- make decisions
- and **take action**

Problem Statement

Build an **Agentic AI System** that transforms **unstructured content into actionable outcomes**.

Your system must:

1. Ingest unstructured input (text, reports, articles, dashboards)
2. Extract key insights
3. Analyze implications
4. Generate recommended actions
5. **Simulate execution of at least one action**
6. Show the resulting system state or outcome

Mandatory Requirement: Google Antigravity

Teams **MUST use Google Antigravity** as the core development platform to:

- orchestrate agent workflows
- manage reasoning and planning
- integrate tools/APIs
- handle execution of actions

Use of additional LLMs is allowed, but **Antigravity must be central to system logic**.

Example Use Cases

Teams may choose any domain such as:

- Business operations
- Public policy
- Logistics / supply chain
- Financial insights
- News analysis
- Urban systems

Example Scenario 1: Business Insight

Input:

A sales report showing declining orders in a region

Output:

Insight:

Orders declined by 25% in Lahore

Impact:

Revenue loss expected

Recommended Action:

Launch regional discount campaign

Simulated Execution:

- Campaign created
- Notification generated

Result:

Projected reach: 5,000 users

Example Scenario 2: Policy / News

Input:

News article about fuel price increase

Output:

Insight:

Fuel prices rising

Impact:

Delivery costs increasing

Recommended Action:

Update delivery pricing

Simulated Execution:

- Pricing table updated
- Customer notification drafted

Result:

New delivery cost applied in checkout simulation

System Requirements

Your solution must include:

1. Content Understanding

- Process unstructured input (text, PDF, website, new article etc.)
- Extract key facts and signals

2. Insight Extraction

- Identify meaningful patterns or insights
- Avoid generic summarization

3. Impact Analysis

- Explain **why the insight matters**
- Connect insight to real-world consequences

4. Action Generation

- Generate **clear, actionable recommendations**
- Must be realistic and domain-relevant

5. Action Simulation (CRITICAL REQUIREMENT)

You must simulate execution of at least **one action**.

Simulation can include:

- Mock API call
- Dashboard update
- CRM / spreadsheet update
- Notification system
- Email/SMS generation
- Workflow trigger

6. Outcome Visualization

Show:

- before vs after state
- logs of action execution
- resulting system change

7. Agentic Workflow

Your system must demonstrate:

- multiple agents OR structured reasoning steps
- planning and execution flow
- traceable decision-making

Deliverables

Teams must submit:

1. Working Prototype with Mobile App (MUST) and Web App (Optional)

2. Demo Video (3–5 minutes)

Must clearly show:

- input → insight → action → simulation → result

3. Agent Trace / Logs from Antigravity

- Workplan
- Tasks Plan
- reasoning steps
- decision flow
- action execution

4. Documentation (README)

Include:

- architecture overview
- tools/APIs used
- how Antigravity is used
- assumptions

Evaluation Criteria

1. Use of Google Antigravity — 25%

- Antigravity used in core orchestration
- Effective handling of reasoning and tool execution
- Not used superficially

2. Agentic Reasoning & Workflow — 20%

- Clear multi-step reasoning
- Logical flow: insight → action → execution
- Evidence of autonomy

3. Insight & Decision Quality — 20%

- Insights are meaningful and non-trivial
- Strong reasoning behind actions
- Outputs are structured and clear

4. Action Simulation & Outcome — 15%

- Action is realistically simulated
- Clear demonstration of system state change
- End-to-end execution visible

5. Technical Implementation — 10%

- Clean architecture
- Tool/API integration
- Robustness and handling of edge cases

6. Innovation & UX — 10%

- Creative approach
- Intuitive interface
- Clear and compelling demo

Important Guidelines

- This is NOT a summarization tool
- Systems must demonstrate **action-oriented intelligence**
- At least one action must be **executed in simulation**
- Use of **real-world inspired scenarios is encouraged**
- No sensitive or real personal data should be used